TECHNICAL DOCUMENTATION

FuseIQ v3.0

AI Agent Command Center — Platform Documentation

VERSION	PHASE	DATE
3.0.0-alpha	2 — Intelligence Layer	April 2026

Abbasi Global Ltd · https://fuseiq.vercel.app · https://github.com/abbasi8586/fuseiq

Table of Contents

01 Project Overview
02 Architecture & Tech Stack
03 Directory Structure
04 Features & Functions
05 Phase 1 Status
06 Phase 2 — Intelligence Layer
07 Design System
08 Database Schema
09 API & Integrations
10 Deployment Guide
11 Future Roadmap

01 — Project Overview

FuseIQ v3.0 is a next-generation AI Agent Orchestration Platform — a command center where humans and AI agents collaborate seamlessly. It provides real-time monitoring, workflow orchestration, cost tracking, and team management for AI-powered operations.

Vision: Build the operating system for AI-native teams — where AI agents are first-class team members with roles, responsibilities, and measurable output.

ATTRIBUTE	VALUE
Owner	Awais Abbasi (Abbasi Global Ltd)
Live URL	https://fuseiq.vercel.app
GitHub	https://github.com/abbasi8586/fuseiq
Version	3.0.0-alpha
License	Proprietary
Current Phase	PHASE 2 — INTELLIGENCE LAYER

02 — Architecture & Tech Stack

LAYER	TECHNOLOGY	VERSION
Framework	Next.js (App Router)	16.2.4
Language	TypeScript	5.x
Styling	Tailwind CSS	v4
UI Components	shadcn/ui + Custom Glass	Latest
Animation	Framer Motion	12.38.0
State Management	Zustand	5.0.12
Backend	Supabase	PostgreSQL + Auth + Realtime
Charts	Recharts	3.8.1
Forms	React Hook Form + Zod	Latest
Icons	Lucide React	Latest
Font	Inter (Google Fonts)	Variable

Key Dependencies

```
{
  "next": "16.2.4",
  "react": "19.2.4",
  "tailwindcss": "^4",
  "framer-motion": "^12.38.0",
  "supabase-js": "^2.104.1",
  "zustand": "^5.0.12",
  "recharts": "^3.8.1"
}
```

03 — Directory Structure

```
fuseiq/
├── .env.local              # Environment variables (gitignored)
├── .env.local.example      # Template for env vars
├── next.config.ts          # Next.js config
├── package.json            # Dependencies
├── tsconfig.json           # TypeScript config
├── components.json         # shadcn/ui config
├── seed-agents.js          # Database seed script
├── test-supabase.js        # Supabase connection test
|
├── public/                 # Static assets
|   ├── images/
|   └── fonts/
|
├── src/
|   ├── app/
|   |   ├── (dashboard)/            # Dashboard routes (grouped)
|   |   |   ├── page.tsx            # Command Center (Dashboard Home)
|   |   |   ├── layout.tsx          # Dashboard shell
|   |   |   ├── staff/              # Staff Directory
|   |   |   ├── agents/             # Agent Forge
|   |   |   ├── operations/         # Operations Center
|   |   |   ├── approvals/          # Approval workflow
|   |   |   ├── analytics/          # Analytics & Reporting
|   |   |   ├── comms/              # Communications hub
|   |   |   ├── swarm/              # Swarm Canvas
|   |   |   ├── simulator/          # Agent Simulator
|   |   |   ├── copilot/            # AI Co-Pilot
|   |   |   ├── marketplace/        # Agent Marketplace
|   |   |   ├── team/               # Team Management
|   |   |   ├── billing/            # Stripe Billing
|   |   |   ├── audit/              # Audit Log
|   |   |   └── settings/           # Settings
|   |   ├── page.tsx                # Landing/Marketing page
|   |   ├── layout.tsx              # Root layout
|   |   └── globals.css             # Global styles + Glassmorphism
|   |
|   ├── components/
|   |   ├── glass/                  # Glassmorphism components
|   |   |   ├── glass-card.tsx
|   |   |   ├── kpi-card.tsx
|   |   |   ├── cost-ticker.tsx
|   |   |   ├── progress-bar.tsx
|   |   |   └── status-badge.tsx
|   |   ├── data-display/
|   |   |   └── activity-feed.tsx
|   |   ├── layout/
|   |   |   ├── sidebar.tsx
|   |   |   └── header.tsx
|   |   └── ui/                     # shadcn/ui primitives
|   |
|   ├── lib/
|   |   ├── supabase/
|   |   |   └── client.ts
|   |   ├── stores/
|   |   |   └── workspace-store.ts
```

```
|   |   └── utils.ts
|   |
|   └── types/
|       └── index.ts
|
└── supabase/
    ├── migrations/                 # SQL migrations
    └── config.toml                 # Supabase CLI config
```

04 — Features & Functions

Command Center (Dashboard)

- **Real-time KPI Cards:** Active Agents, Executions Today, Success Rate, Avg Cost/Run
- **Active Agents List:** Live status, efficiency scores, execution counts
- **Execution Trends:** Time-series bar chart (1H, 24H, 7D, 30D)
- **Pending Approvals:** Risk-based approval cards with Approve/Reject
- **Activity Feed:** Real-time stream of agent events
- **Cost Ticker:** Live cost breakdown by provider

Staff Directory

- Server-side data fetching from Supabase `agents` table
- Grid/table view of all AI and Human team members
- Filters by department, status, type (AI/Human)
- Efficiency scores and workload metrics
- Role-based organization

Phase 2 Features (In Progress)

- **Agent Forge:** Create and configure AI agents with framework selection
- **Operations Center:** Kanban board for task management
- **Approval Workflow:** Risk-based routing with audit trail
- **Analytics:** Execution volume, cost analysis, efficiency heatmaps
- **Communications Hub:** Team chat channels (AI + Human)
- **Swarm Canvas:** Visual workflow builder (drag-and-drop)
- **AI Co-Pilot:** Command palette (⌘K) with natural language
- **Marketplace:** Pre-built agent templates
- **Billing & Usage:** Stripe subscription management
- **Audit Log:** Immutable activity history

05 — Phase 1 Status (Completed)

FEATURE	STATUS	NOTES
Project scaffold	✅ Complete	Next.js 16 + Tailwind v4 + shadcn/ui
Glassmorphism design system	✅ Complete	Custom CSS utilities, glass-card, neon-button
Sidebar navigation	✅ Complete	15 pages, collapsible, mode toggle
Header with search	✅ Complete	Command palette trigger, notifications
Command Center dashboard	✅ Complete	KPIs, agent list, charts, approvals, activity feed
Supabase integration	✅ Complete	Real env vars, working client
Database schema	✅ Complete	18 tables created
Agent seeding	✅ Complete	8 AI agents pre-populated
Staff Directory	✅ Complete	Server + Client, live Supabase data
Vercel deployment	✅ Complete	Live at https://fuseiq.vercel.app

06 — Phase 2 — Intelligence Layer (May 2026)

Week 1-2: Authentication & Security

- Supabase Auth integration (email/password, OAuth)
- Role-based access control (Director, Manager, Member, Viewer)
- Workspace/tenant isolation
- API key encryption at rest
- Row Level Security (RLS) policies

Week 3-4: Real-time Data

- Supabase Realtime subscriptions
- Live agent status updates
- Activity feed streaming
- Cost ticker with live updates
- Notification system (in-app + email)

Week 5-6: Co-Pilot & Command Palette

- ⌘K command palette component
- Natural language parser
- Quick actions (deploy agent, create task, view analytics)
- Context-aware suggestions
- Voice command support (Web Speech API)

Week 7-8: Swarm Canvas MVP

- React Flow canvas component
- Drag-and-drop node builder
- Agent node, Task node, Condition node, Trigger node
- Simple sequential workflows
- Workflow execution engine (edge function)

Week 9-10: Agent Forge Completion

- Agent creation wizard (5-step form)
- Framework-specific configuration
- API key management (encrypted)
- Testing console (live agent test)
- Agent versioning

07 — Design System

Color Palette (Cyber-Luxury Glassmorphism)

TOKEN	HEX	USAGE
Void	#06070A	Primary background
Abyss	#0A0B10	Sidebar, panels
Depth	#0F111A	Cards, elevated surfaces
Surface	#161925	Glass card background (60% opacity)
Ridge	#1E2233	Borders, dividers
Shard	#252A3D	Hover states
Primary Cyan	#00D4FF	CTAs, active states, AI indicators
Aurora	#B829DD	Secondary accent, gradients
Ember	#FF6B35	Human mode, warnings
Signal	#00E5A0	Success, online status
Warn	#FFC857	Medium risk, pending
Danger	#FF4757	Errors, rejections, offline
Text Hero	#FFFFFF	Headlines
Text Body	#B8BED8	Body text
Text Muted	#6B7290	Secondary text
Text Faint	#4A5068	Borders, disabled

Glassmorphism Levels

```
/* Level 1: Cards */
.glass-card {
  background: rgba(22, 25, 37, 0.6);
  backdrop-filter: blur(20px);
  border: 1px solid rgba(255, 255, 255, 0.08);
  border-radius: 16px;
}

/* Level 2: Panels (header, sidebar) */
.glass-panel {
  background: rgba(10, 11, 16, 0.8);
  backdrop-filter: blur(12px);
  border: 1px solid rgba(255, 255, 255, 0.06);
}

/* Level 3: Inputs */
.glass-input {
  background: rgba(255, 255, 255, 0.03);
  border: 1px solid #1E2233;
  border-radius: 10px;
}
```

Typography

ELEMENT	FONT	WEIGHT	SIZE
Page Title	Inter	700	24px
Section Header	Inter	600	18px
Card Title	Inter	500	16px
Body	Inter	400	14px
Caption	Inter	400	12px
Micro	Inter	500	10px

Spacing System

- Base unit: 4px
- Card padding: 20px (5 units)
- Section gap: 24px (6 units)
- Grid gap: 16px (4 units)

Animations

ANIMATION	DURATION	EASING
Card hover	300ms	ease-out
Page transition	300ms	ease-out
Stagger children	50ms each	ease-out
Chart bars	500ms	ease-out
Glow pulse	2s infinite	ease-in-out

08 — Database Schema (Supabase)

Core Tables

TABLE	PURPOSE	ROWS (EST.)
`agents`	AI agent definitions	50+
`staff`	Human team members	20+
`tasks`	Operations tasks	500+
`approvals`	Approval workflow items	100+
`providers`	AI provider configs	10
`workflows`	Swarm workflow definitions	25
`executions`	Agent run history	10K+
`activity_logs`	Audit trail	50K+
`messages`	Communications	5K+
`channels`	Chat channels	20
`billing_usage`	Metered usage	100K+
`subscriptions`	Stripe subscriptions	50
`workspaces`	Multi-tenant workspaces	10

Key Relationships

```
workspaces (1) → agents (N)
workspaces (1) → staff (N)
workspaces (1) → tasks (N)
agents (1) → executions (N)
agents (1) → approvals (N)
staff (1) → tasks (N) [as assignee]
workflows (1) → executions (N)
```

09 — API & Integrations

Supabase APIs

FEATURE	API	STATUS
Agent CRUD	REST/PostgREST	✅ Active
Staff Directory	REST + Server Components	✅ Active
Real-time updates	Supabase Realtime	🚧 Phase 2
Auth	Supabase Auth	🚧 Phase 2
Storage	Supabase Storage	⬜ Planned
Edge Functions	Deno/Edge Runtime	⬜ Planned

External Integrations (Planned)

SERVICE	PURPOSE
Stripe	Billing & subscriptions
OpenAI	GPT-4, GPT-3.5 agents
Kimi API	Claude agent backend
GitHub	Code review agents, PR automation
Slack	Notifications, bot integration
Airtable	Data sync agents
Zapier	Workflow automation
SendGrid	Email notifications

10 — Deployment Guide

Vercel (Production)

```
# Deploy latest
vercel --prod

# Environment variables required:
# NEXT_PUBLIC_SUPABASE_URL
# NEXT_PUBLIC_SUPABASE_ANON_KEY
# SUPABASE_SERVICE_ROLE_KEY
# STRIPE_PUBLISHABLE_KEY (Phase 2)
# STRIPE_SECRET_KEY (Phase 2)
```

Local Development

```
cd /Users/hyp3r/.openclaw/workspace/fuseiq
npm install
npm run dev
# Open http://localhost:3000
```

Database Management

```
# Push schema changes
supabase db push

# Seed data
node seed-agents.js

# Generate types
supabase gen types typescript --project-id kowwqfadifawowfrdiyz --schema public > src/types/supab
```

11 — Future Roadmap

Phase 3 — Scale & Marketplace (June 2026)

- Agent Marketplace (browse, install, rate)
- Pre-built workflow templates
- Team collaboration features
- Advanced analytics (custom reports, export)
- Mobile-responsive layout
- Dark/light mode toggle
- Public API with rate limiting
- Webhook integrations (Zapier, Make)

Phase 4 — Enterprise (July 2026)

- SAML/SSO integration
- Audit log compliance (SOC 2)
- White-label customization
- On-premise deployment option
- Advanced security (IP allowlisting, MFA)
- Custom integrations SDK